

RAJALAKSHMI ENGINEERING COLLEGE

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Web Technology And Mobile Application

Experiment 7

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By

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AIM:

To develop a Java Servlet-based module for an **Online Railway Reservation System** that stores user preferences such as theme and language using cookies, and retrieves these preferences to personalize the interface on subsequent visits.

ALGORITHM:

1. Start

2. Handle Client Request (GET Method)

- Set response content type to HTML
- Initialize default values:
 - Theme = "Light"
 - Language = "English"

3. Retrieve Cookies

- Get all cookies from the request
- If cookies exist:
 - Iterate through each cookie
 - If cookie name = "theme", store its value
 - If cookie name = "language", store its value

4. Display Web Page

- Show Railway Reservation System title
- Display stored (or default) preferences
- Provide a form for user to select:
 - Theme (Light/Dark)
 - Language (English/Hindi/Tamil)

5. Handle Form Submission (POST Method)

- Read user input values (theme, language)
- Create cookies for theme and language
- Set cookie expiry time (e.g., 1 day)
- Add cookies to response

6. Redirect Request

- Redirect to the same servlet (GET method)

7. Display Updated Preferences

- Retrieved cookies are shown as user preferences

CODE:

Servlet code:

```
package wtma;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.io.IOException;
import java.io.PrintWriter;

@WebServlet("/railwayPreferences")
public class exp7 extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String theme = "Light";
        String language = "English";

        // Retrieve cookies
        Cookie[] cookies = request.getCookies();
        if (cookies != null) {
            for (Cookie c : cookies) {
                if (c.getName().equals("theme")) theme = c.getValue();
                if (c.getName().equals("language")) language = c.getValue();
            }
        }

        String cardColor = theme.equals("Dark") ? "#1e1e1e" : "#ffffff";
        String textColor = theme.equals("Dark") ? "#ffffff" : "#000000";

        out.println("<html><head><title>Railway Preferences</title>");
        out.println("<style>");

        out.println("body { margin:0; font-family: 'Segoe UI'; background-image: url('background orrs.avif'); color:" + textColor + "; background-size: cover;background-position: center;background-repeat: no-repeat;background-attachment: fixed;background-color: whitesmoke; }");

        out.println(".container { width: 400px; margin: 80px auto; background:" + cardColor + "; padding: 30px; border-radius: 15px; box-shadow: 0 8px 20px rgba(0,0,0,0.2); text-align:center; }");

        out.println("h1 { margin-bottom: 10px; }");
        out.println("h2 { color: #4facfe; }");
```

```

out.println("select, input[type=submit] { width: 80%; padding: 10px; margin: 10px 0; border-radius: 8px; border: none; font-size: 15px; }");

out.println("input[type=submit] { background: #4facfe; color: white; cursor: pointer; transition: 0.3s; }");
out.println("input[type=submit]:hover { background: #007bff; }");

out.println(".pref { margin: 10px 0; font-size: 16px; }");

out.println("</style></head><body>");

out.println("<div class='container'>");

out.println("<h1> Railway Reservation</h1>");
out.println("<h2>User Preferences</h2>");

out.println("<div class='pref'><b>Theme:</b> " + theme + "</div>");
out.println("<div class='pref'><b>Language:</b> " + language + "</div>");

out.println("<form method='post'>");

out.println("<select name='theme'>");
out.println("<option value='Light'> Light</option>");
out.println("<option value='Dark'> Dark</option>");
out.println("</select>");

out.println("<select name='language'>");
out.println("<option value='English'>English</option>");
out.println("<option value='Hindi'>Hindi</option>");
out.println("<option value='Tamil'>Tamil</option>");
out.println("</select>");

out.println("<input type='submit' value=' Save Preferences'>");

out.println("</form>");
out.println("</div>");

out.println("</body></html>");
}

protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

String theme = request.getParameter("theme");
String language = request.getParameter("language");

Cookie themeCookie = new Cookie("theme", theme);
Cookie languageCookie = new Cookie("language", language);

themeCookie.setMaxAge(24 * 60 * 60);
languageCookie.setMaxAge(24 * 60 * 60);

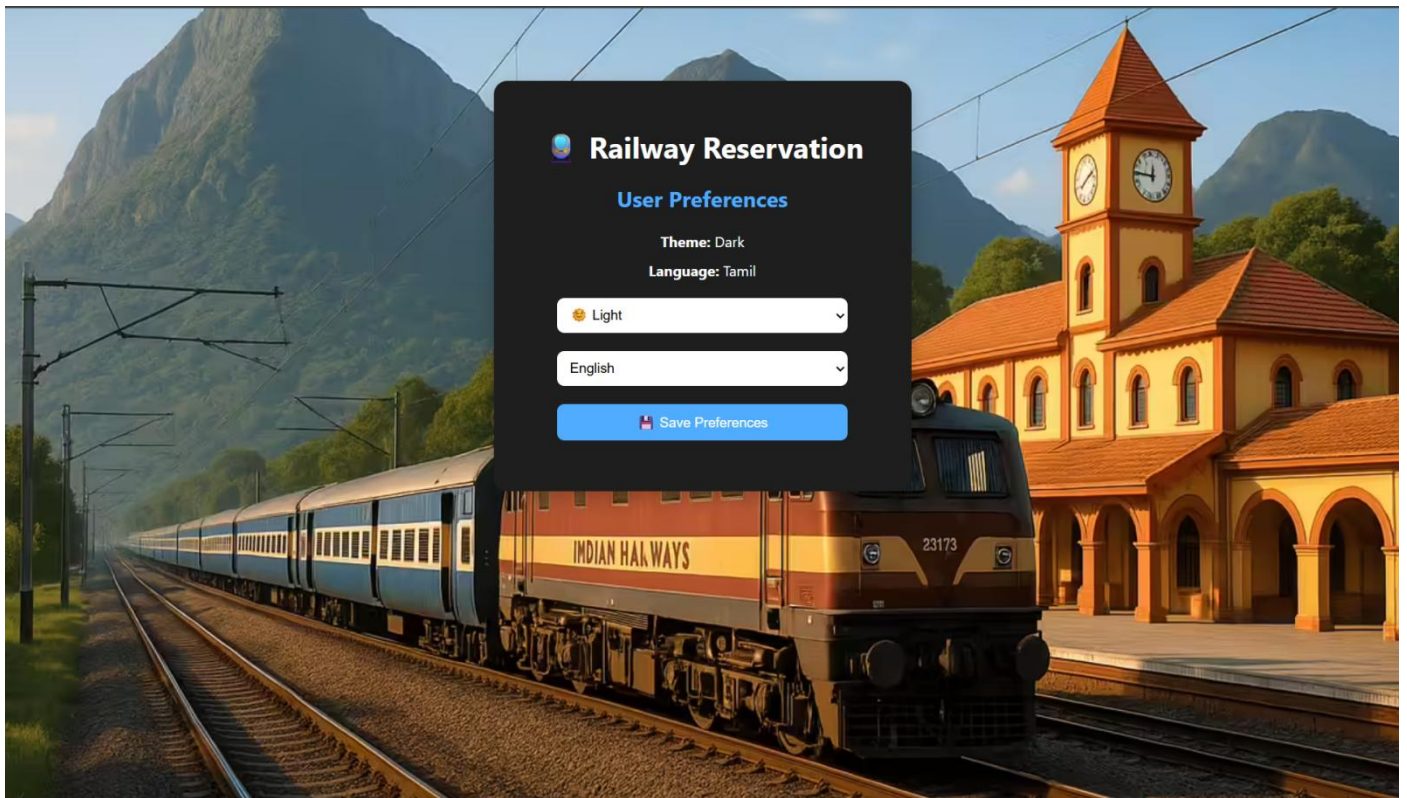
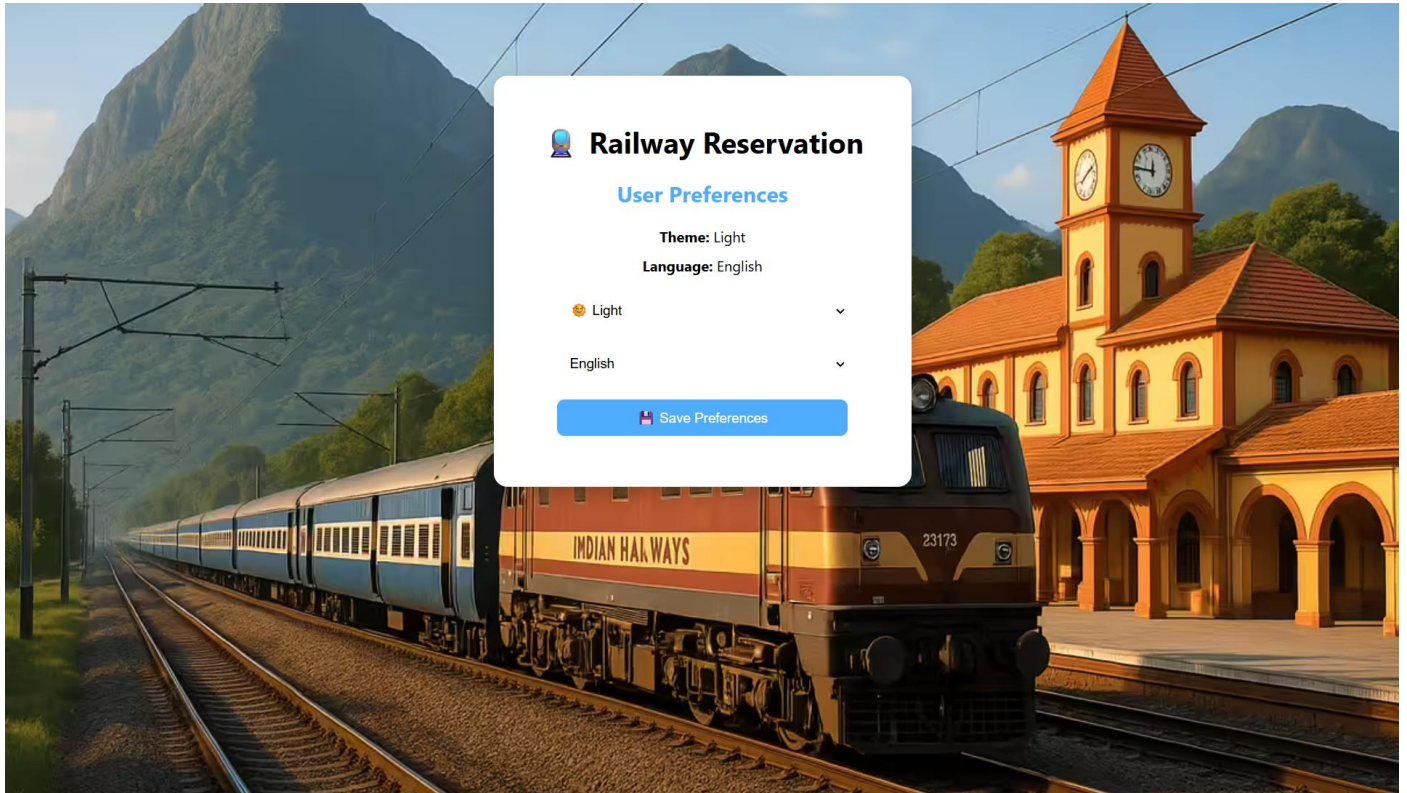
response.addCookie(themeCookie);
response.addCookie(languageCookie);

```

```
response.sendRedirect("railwayPreferences");
```

```
}
```

OUTPUT:



RESULT:

The servlet successfully stores user preferences (theme and language) using cookies and retrieves them on subsequent visits, enabling personalized display in the online railway reservation system.